

26021 Kemi (Polyteknisk grundlag)/26021 Chemistry (Polytechnic Foundation) non- random version - Spring 2025

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:

Der er altid netop ét svar som er mere rigtigt end de andre

Studerende kan kun vælge ét svar per spørgsmål

Hvert rigtigt svar giver 1 point

Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

The following approach to scoring responses is implemented and is based on "One best answer"

There is always only one correct answer – a response that is more correct than the rest

Students are only able to select one answer per question

Every correct answer corresponds to 1 point

Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

DK: Hvad er den omtrentlige forventede H-N-H bindingsvinkel i N_2H_4 (hydrazin)?

EN: *What is the approximate expected H-N-H bond angle in N_2H_4 (hydrazine)?*

Choose one answer

☐ 107°

☐ 90°

☐ 120°

☐ 180°

☐ 54°

DK: Hvor mange mol ioner dannes ved komplet opløsning af 1 mol aluminiumsulfat i vand?

EN: *How many moles of ions are formed by complete dissolution of 1 mol of aluminum sulfate in water?*

Choose one answer

- ☐ 5 mol
- ☐ 4 mol
- ☐ 2 mol
- ☐ 6 mol

DK: Hvilket af følgende molekyler har ikke en plan geometri?

EN: Which of the following molecules does not possess a planar geometry?

Choose one answer

☐ SO_3

☐ SiH_4

☐ BH_3

☐ SO_2

☐ C_2H_4

DK: Hvilket udsagn er korrekt om en katalysators påvirkning på en kemisk reaktion?

EN: Which statement is correct for a catalyst's influence on a chemical reaction?

Choose one answer

- ☐ Den sænker aktiveringsenergien. *It lowers the activation energy.*
- ☐ Den bliver forbrugt under reaktionen. *It is consumed during the reaction.*
- ☐ Den hæver aktiveringsenergien. *It raises the activation energy.*
- ☐ Den forskyder reaktionslignevægten mod produkterne. *It shifts the reaction equilibrium towards the products.*
- ☐ Den forskyder reaktionslignevægten mod reaktanterne. *It shifts the reaction equilibrium towards the reactants.*

DK: Hvad er oxidationstallet for svovl i svovlsyre, H_2SO_4 ?

EN: *What is the oxidation state for sulfur in sulfuric acid, H_2SO_4 ?*

Choose one answer

☐ +IV (+4)

☐ +VI (+6)

☐ -II (-2)

☐ 0

☐ +II (+2)

DK: Hvilken af følgende forbindelser har det højeste damptryk ved stuetemperatur?

EN: Which of the following possess the largest vapor pressure at room temperature?

Choose one answer

- ☐ Vand. *Water.*
- ☐ Glucose. *Glucose.*
- ☐ Natriumchlorid. *Sodium chloride.*
- ☐ Ethanol. *Ethanol.*

DK: Hvor mange alkoholer har molekylformlen $C_4H_{10}O$?

EN: *How many alcohols have the molecular formula $C_4H_{10}O$?*

Choose one answer

☐ 3

☐ 4

☐ 1

☐ 2

☐ 5

☐ 6

DK: Hvor mange elektroner har fosfor i sin yderste skal?

EN: *How many electrons does phosphorous possess in its outermost shell?*

Choose one answer

☐ 6

☐ 5

☐ 4

☐ 3

☐ 7

DK: Hvilket udsagn er korrekt om CS₂ (som har kulstof som centralatom)?

EN: Which statement is correct for CS₂ (which has carbon as a central atom)?

Choose one answer

- ☐ Molekylet er vinklet og polært. *The molecule is bent and polar.*
- ☐ Molekylet er vinklet og upolært. *The molecule is bent and non-polar.*
- ☐ Molekylet er lineært og upolært. *The molecule is linear and non-polar.*
- ☐ Molekylet er lineært og polært. *The molecule is linear and polar.*

DK: Et galvanisk element konstrueres med en halvcelle bestående af $\text{Fe}^{3+}/\text{Fe}^{2+}$ ($E^\circ = +0.77 \text{ V}$) og en halvcelle bestående af $\text{Ce}^{4+}/\text{Ce}^{3+}$ ($E^\circ = +1.6 \text{ V}$). Hvilket af følgende udsagn er korrekt?

EN: A galvanic element is constructed with a half-cell consisting of $\text{Fe}^{3+}/\text{Fe}^{2+}$ ($E^\circ = +0.77 \text{ V}$) and a half-cell consisting of $\text{Ce}^{4+}/\text{Ce}^{3+}$ ($E^\circ = +1.6 \text{ V}$). Which of the following statements is true?

Choose one answer

- ☐ Elektroner strømmer fra Ce^{3+} til Fe^{3+} . *Elektrons are flowing from Ce^{3+} to Fe^{3+}*
- ☐ Fe^{3+} oxideres til Fe^{2+} . *Fe^{3+} is being oxidized to Fe^{2+} .*
- ☐ Fe^{2+} reduceres til Fe^{3+} . *Fe^{2+} is being reduced to Fe^{3+} .*
- ☐ Ce^{4+} reduceres til Ce^{3+} . *Ce^{4+} is being reduced to Ce^{3+}*
- ☐ Ce^{4+} oxideres til Ce^{3+} . *Ce^{4+} is being oxidized to Ce^{3+} .*

DK: En given kemisk reaktion har $\Delta H = -50 \text{ kJ/mol}$ og $\Delta S = -100 \text{ J/(mol K)}$. Ved hvilken temperatur skifter reaktionen fra spontan til ikke-spontan?

EN: A given chemical reaction has $\Delta H = -50 \text{ kJ/mol}$ and $\Delta S = -100 \text{ J/(mol K)}$. At which temperature does the reaction shift from spontaneous to non-spontaneous?

Choose one answer

- ☐ 298 K
- ☐ 600 K
- ☐ 500 K
- ☐ 400 K
- ☐ 273 K

DK: En opløsning indeholder både Pb^{2+} og Ag^+ . Ved $[\text{Cl}^-] = 1.0 \times 10^{-3} \text{ M}$, hvilket af saltene PbCl_2 og AgCl vil udfælde først, dvs. ved den lavest koncentration af metal-ionen?

De relevante opløselighedsprodukter er $K_{\text{sp}}(\text{AgCl}) = 1.8 \times 10^{-10} \text{ M}^2$ og $K_{\text{sp}}(\text{PbCl}_2) = 1.7 \times 10^{-5} \text{ M}^3$.

EN: A solution contains both Pb^{2+} and Ag^+ . At $[\text{Cl}^-] = 1.0 \times 10^{-3} \text{ M}$, which of the salts, PbCl_2 and AgCl , will start to precipitate first, i.e. at the lowest concentration of the metal ion?

The relevant solubility products are $K_{\text{sp}}(\text{AgCl}) = 1.8 \times 10^{-10} \text{ M}^2$ and $K_{\text{sp}}(\text{PbCl}_2) = 1.7 \times 10^{-5} \text{ M}^3$.

Choose one answer

- ☐ Begge samtidig. *Both at the same time.*
- ☐ Kan ikke afgøres. *Cannot be determined.*
- ☐ AgCl
- ☐ PbCl_2

DK: Hvad er den formelle ladning på nitrogenatomet i nitrat (NO_3^-) bestemt ud fra dens Lewis-struktur?

EN: *What is the formal charge on the nitrogen atom in nitrate (NO_3^-) as determined from its Lewis structure?*

Choose one answer

☐ +2

☐ 0

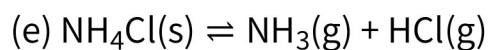
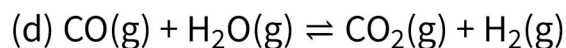
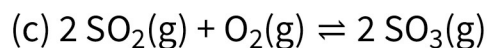
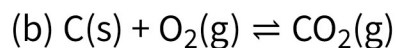
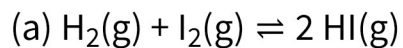
☐ 1+

☐ -1

☐ -2

DK: For hvilken af følgende reaktioner vil en stigning i trykket favorisere dannelsen af produkterne? (se reaktioner under engelsk tekst).

EN: For which of the following reactions would an increase in the pressure favor the formation of the products.



Choose one answer

☐ (b)

☐ (d)

☐ (c)

☐ (a)

☐ (e)

DK: Hvilken molekylær egenskab forklarer det relativt høje kogepunkt for HF i forhold til HCl, HBr og HI?

EN: Which molecular property explains the relatively high boiling point for HF in comparison to HCl, HBr, and HI?

Choose one answer

- ☐ Den molare masse. *The molar mass.*
- ☐ Kovalent binding. *Covalent bonding*
- ☐ Ion-dipol-vekselvirkninger. *Ion-dipole interactions*
- ☐ Ion-binding. *Ionic bonding*
- ☐ Hydrogenbinding. *Hydrogen bonding*
- ☐ van der Waals-vekselvirkninger. *van der Waals interactions.*

DK: Hvad er pH i en 0.10 M vandig opløsning af eddikesyre (CH_3COOH , $\text{p}K_a = 4.7$).

EN: What is the pH in a 0.1M aqueous solution of acetic acid (CH_3COOH , $\text{p}K_a = 4.7$).

Choose one answer

☐ 7.0

☐ 4.7

☐ 2.9

☐ 5.2

☐ 1.0

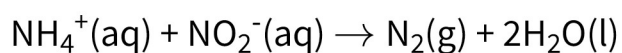
DK: En buffer-opløsning blev fremstillet ved at blande 1.00 mol NH_3 ($K_b = 1.80 \times 10^{-5}$ M) med 1.00 mol NH_4Cl i vand med et endeligt totalvolumen på 1.00 L. Til 500 mL af denne opløsning blev der tilføjet 30.0 mL af en 1.00 M NaOH(aq) -opløsning. Hvad blev pH af denne opløsning?

EN: A buffer solution was prepared by mixing 1.00 mol NH_3 ($K_b = 1.80 \times 10^{-5}$ M) with 1.00 mol NH_4Cl in water with a final total volume of 1.00 L. To 500 mL of this solution was added 30.0 mL of a 1.00 M NaOH(aq) solution. What is the resulting pH of this solution?

Choose one answer

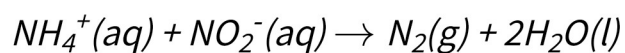
- ☐ 9.31
- ☐ 8.96
- ☐ 9.83
- ☐ 9.57

DK: Hvad er hastighedsudtrykket, $k [\text{NH}_4^+]^x [\text{NO}_2^-]^y$ (hvor k er en konstant) for reaktionen:



hvor følgende data er blevet bestemt (under engelsk tekst):

EN: What is the rate law, $k [\text{NH}_4^+]^x [\text{NO}_2^-]^y$ (where k is a constant) for the reaction:



where the following data have been determined:

Experiment number	$[\text{NH}_4^+]$ (M)	$[\text{NO}_2^-]$ (M)	Reaction rate (M/s)
1	0.010	0.020	0.020
2	0.015	0.020	0.030
3	0.010	0.010	0.005

Choose one answer

- ☐ $k [\text{NH}_4^+][\text{NO}_2^-]^2$
- ☐ $k [\text{NH}_4^+]^2[\text{NO}_2^-]$
- ☐ $k [\text{NH}_4^+]^2[\text{NO}_2^-]^2$
- ☐ Ingen af disse
- ☐ $k [\text{NH}_4^+][\text{NO}_2^-]$

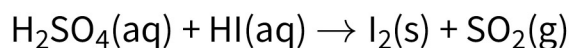
DK: Hvilken af følgende ioner har den mindste ionradius?

EN: Which of the following ions have the smallest ionic radius?

Choose one answer

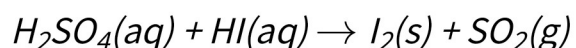
- ☐ Rb^+
- ☐ Sr^{2+}
- ☐ Br^-
- ☐ Se^{2-}
- ☐ De har alle samme størrelse da de har samme antal elektroner.

DK: Afstem følgende reaktion for oxidationen af HI(aq) med svovlsyre.



Hvad er summen af koefficienterne? Brug de laveste mulige, heltallige koefficienter. Glem ikke koefficienter af en (1).

EN: Balance the following reaction for the oxidation of HI(aq) with sulfuric acid:



What is the sum of the coefficients? Use the smallest possible, integer coefficients. Do not forget coefficients of one (1).

Choose one answer

☐ 7

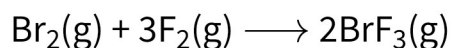
☐ 9

☐ 5

☐ 11

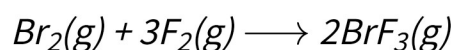
☐ 13

DK: Estimer ΔH for følgende reaktion



givet bindingsenthalpierne for Br-Br (-192 kJ/mol), F-F (-158 kJ/mol) og Br-F (-197 kJ/mol).

EN: Estimate ΔH for the following reaction

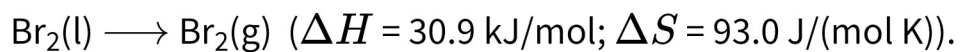


given the bond enthalpies for Br-Br (-192 kJ/mol), F-F (-158 kJ/mol) and Br-F (-197 kJ/mol).

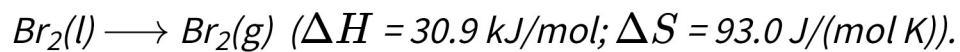
Choose one answer

- ☐ -611 kJ/mol
- ☐ -410 kJ/mol
- ☐ -516 kJ/mol
- ☐ -665 kJ/mol
- ☐ -720 kJ/mol

DK: Estimer kogepunktet for Br₂.



EN: Estimate the boiling point for Br₂.



Choose one answer

- ☐ 59°C
- ☐ 85°C
- ☐ 44°C
- ☐ 373°C

DK: Hvilket af følgende udsagn er korrekt for en 0.10 M opløsning af en svag syre, HX?

EN: Which of the following statements is correct for a 0.10 M solution of a weak acid, HX?

Choose one answer

- ☐ $[HX] > [H^+]$
- ☐ $pH = 1$
- ☐ $pH = 1, [H^+] = 0.10 \text{ M}$
- ☐ $[H^+] = 0.10 \text{ M}$
- ☐ $[X^-] = 0.10 \text{ M}$

DK: Hvad er den molære opløselighed af $\text{Cu}(\text{OH})_2$ i vand givet at $K_{\text{sp}}(\text{Cu}(\text{OH})_2) = 1.6 \times 10^{-19} \text{ M}^3$?

EN: What is the molar solubility of $\text{Cu}(\text{OH})_2$ in water given that $K_{\text{sp}}(\text{Cu}(\text{OH})_2) = 1.6 \times 10^{-19} \text{ M}^3$?

Choose one answer

- ☐ $6.3 \times 10^{-10} \text{ M}$
- ☐ $3.4 \times 10^{-7} \text{ M}$
- ☐ $2.0 \times 10^{-5} \text{ M}$
- ☐ $1.6 \times 10^{-19} \text{ M}$
- ☐ $2.4 \times 10^{-20} \text{ M}$

DK: Atomet med elektronkonfigurationen $4s^24p^5$ i yderste skal vil være i:

EN: *The atom with the electronic configuration $4s^24p^5$ in the outermost shell will be in:*

Choose one answer

- ☐ 4. hovedgruppe, 7. periode. 4^{th} main group, 7^{th} period.
- ☐ 7. hovedgruppe, 4. periode. 7^{th} main group, 4^{th} period.
- ☐ 7. hovedgruppe, 5. periode. 7^{th} main group, 5^{th} period.
- ☐ 5. hovedgruppe, 4. periode. 5^{th} main group, 4^{th} period.
- ☐ 4. hovedgruppe, 5. periode. 4^{th} main group, 5^{th} period.

DK: En blanding af 0.50 mol $\text{H}_2(\text{g})$ og 0.50 mol $\text{N}_2(\text{g})$ fyldes i en 25 L-beholder ved 30°C . Beholderen punkteres med en nål. Efter noget tid, hvilket af følgende udsagn er korrekt?

EN: A mixture of 0.50 mol $\text{H}_2(\text{g})$ and 0.50 mol $\text{N}_2(\text{g})$ are placed in a 25 L container at 30°C . The container is punctured with a needle. After some time, which of the following statements is correct?

Choose one answer

- ☐ Kun partialtrykket af H_2 er steget. *Only the partial pressure of H_2 has increased.*
- ☐ Partialtrykkene af de to gasser er identisk. *The partial pressures of the gases are identical.*
- ☐ Partialtrykket af N_2 er større end partialtrykket af H_2 . *The partial pressure of N_2 is larger than the partial pressure of H_2 .*
- ☐ Partialtrykket af H_2 er større end partialtrykket af N_2 . *The partial pressure of H_2 is larger than the partial pressure of N_2 .*
- ☐ Partialtrykkene af begge gasser er forøget. *The partial pressures of both gases have increased?*

DK: En opløsning blev fremstillet ved at opløse 9.81 g af et ukendt molekyle (ikke-flygtig og ikke-elektrolyt) i 90.0 g vand. Kogepunktet for opløsningen blev bestemt til 100.37°C ved 1.00 atm. Hvad er den omtrentlige molare masse af det ukendte molekyle? (For vand er $K_b = 0.51^{\circ}\text{C/m}$).

EN: *As solution was prepared by dissolving 9.81 g of an unknown molecule (non-volatile and non-electrolyte) in 90.0 g of water. The boiling point of the solution was determined to be 100.37°C at 1.00 atm. what is the approximate molar mass of the unknown molecule? ($K_b = 0.51^{\circ}\text{C/m}$ for water).*

Choose one answer

- ☐ 480 g/mol
- ☐ 150 g/mol
- ☐ 300 g/mol
- ☐ 52 g/mol
- ☐ 18 g/mol

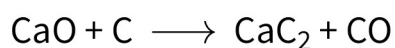
DK: Betragt følgende: ^{72}Zn , ^{75}As , ^{74}Ge . Disse atomer har:

EN: Consider the following: ^{72}Zn , ^{75}As , ^{74}Ge . These atoms have:

Choose one answer

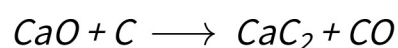
- ☐ Det samme antal neutroner. *The same number of neutrons.*
- ☐ Det samme antal protoner. *The same number of protons.*
- ☐ Det samme antal elektroner. *The same number of electrons.*
- ☐ Det samme antal protoner og neutroner. *The same number of protons and neutrons.*
- ☐ Det samme nukleontal (mass number). *The same mass number.*

DK: Calciumcarbid, CaC_2 , er et vigtigt industrielt kemikalie der fremstillet ved at opvarme calciumoxid med kulstof jævnt før følgende, ikke-afstemte reaktionsskema:



Hvor meget CaC_2 kan der maksimalt produceres fra reaktionen af CaO (i overskud) med 10.2 g kulstof (C)?

EN: Calcium carbide, CaC_2 , is an important industrial chemical that is prepared by heating CaO with carbon according to the following, non-balanced reaction scheme:

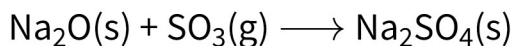


What is the maximum amount of CaC_2 that can be produced from the reaction of CaO (in excess) with 10.2 g carbon (C)?

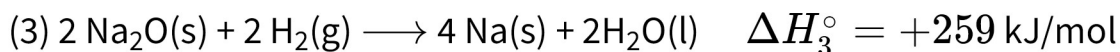
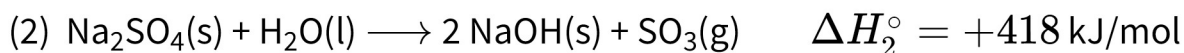
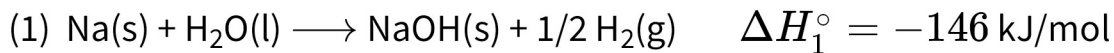
Choose one answer

- ☐ 6.0 g
- ☐ 18.1 g
- ☐ 38.4 g
- ☐ 54 g
- ☐ 27.2 g

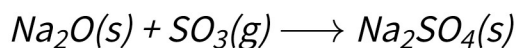
DK: Udregn ΔH° for reaktionen:



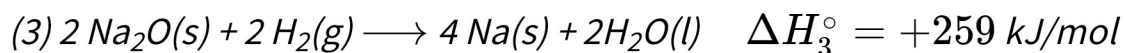
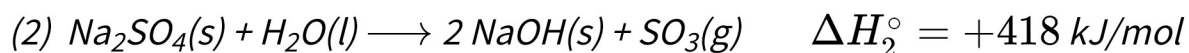
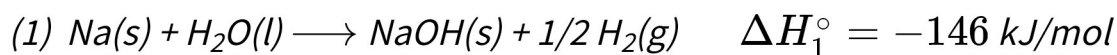
givet følgende information:



EN: Calculate ΔH° for the reaction



given the following information:



Choose one answer

- ☐ -710 kJ/mol
- ☐ -581 kJ/mol
- ☐ 581 kJ/mol
- ☐ 710 kJ/mol
- ☐ -451 kJ/mol
- ☐ 451 kJ/mol
- ☐ +387 kJ/mol
- ☐ -387 kJ/mol

